

Aidan Burrowes

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Education

University of Florida, Gainesville, FL

Expected May 2026

Bachelor of Science in Computer Science, Minors in Electrical Engineering & Physics

GPA 3.88/4.0, UF Honors Program

Relevant Coursework: Operating Systems, Networks, Databases, Intro ML, Data Structures & Algorithms

Work Experience

Roblox, San Mateo, CA

Campus Ambassador

Aug 2025 – Present

- Official representative for Roblox at UF; promoting early career programs with monthly events & \$5,000 budget

Software Engineer Intern

May 2025 – Aug 2025

- Developed automated CI break detection algorithm in Python, **increasing CI efficiency by 50% (3h faster)**
- Created dynamic, automated TeamCity configs. for CI builds, **reducing developer life cycle from 24h to 5m**
- Built scalable full stack app—React/TS, Go microservice, CockroachDB, Docker/Nomad, Grafana/Kibana; **improved productivity by ~20% across 250+ engineers** via real-time build analytics & new query language

Microsoft, Redmond, WA

May 2024 – Aug 2024

Software Engineering Intern

- Enhanced Python tool, utilizing LLMs/TTS to generate synthetic Teams meeting data for 150 languages
- Increased tool efficiency & consistency; scaled audio output by 400% & reduced hallucinations to under 2%
- Configured CI/CD pipelines via Azure DevOps, **generating 20 datasets** for training, testing, & validation
- Fine-tuned Azure STT models, **increasing entity transcription accuracy by 50%** in Teams meetings

HumanAI, Singapore, Singapore

May 2023 – Jul 2023

Software Engineer Intern

- Built Flutter mobile app & REST APIs for iOS and Android; Node.js, Express.js, MongoDB, & robotic chef
- Optimized data pipeline via caching & lazy loading, **achieving 100% faster application loading times**
- Engineered new ingredient-drop mechanism using CAD, **reducing dispensing latency from 15s to 5s**

Research Experience

UF Intelligent Clinical Care Center, Gainesville, FL

Jun 2025 – Present

Machine Learning Engineer Intern

- Developing AI agents via MCP & RAG, automating 5 HR workflows & **reducing work by 10 hours/week**

Data-Driven Humanities Research Group, Gainesville, FL

Jan 2023 – Present

Undergraduate Research Assistant

- Secured competitive AI Scholar designation (1 of 35); **\$1,750 funding** to lead ancient Latin & Greek AI research
- Led 5-person team to build lemmatizers in Python, parsing 2M+ tokens & **increasing dataset by 50%**
- Fine-tuned LLMs on Latin syntax via TensorFlow, achieving up to **98% accuracy** on classification tasks

Gator Glaciology, Gainesville, FL

Aug 2023 – May 2024

Undergraduate Research Assistant

- Optimized GStatSim, a Python geostatistics package, using PyTorch, GPU acceleration, & vectorization
- Expedited interpolation of Antarctic/Greenlandic bed topography maps, achieving a **4x speedup (8m to 2m)**

Projects

United States Air Force, Eglin Air Force Base, FL

Aug 2024 – Dec 2024

- Designed AI-driven, modular architecture for autonomous agents, supporting USAF mission planning
- Interviewed 40+ stakeholders & developed MVP using NVIDIA IsaacSim with multi-agent RL behavior

Publications & Professional Service

- Coauthor**, 2D spintronics for neuromorphic computing, *JLPEA*, doi.org/10.3390/jlpea15020016 Mar 2025
- Coauthor**, sentiment analysis for ancient Greek literature, *DSH*, doi.org/10.1093/dsh/fqae048 Sep 2024
- Author**, syntactic analysis of ancient Latin literature using AI, *UF JUR*, [IPaper presentation!](#) May 2024
- Reviewer**, Oxford University Press, *Digital Scholarship in the Humanities* Jan 2025 – Present

Skills

Languages | Python, Java, Golang, C++, TypeScript, JavaScript, SQL, MATLAB, Dart, HTML/CSS

Technologies | PyTorch, TensorFlow, AWS, Docker, Nomad, Grafana, Kibana, MongoDB, Linux, Git, React, Node.js